Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/01/2025
PI Name:	Oubre, Cherie Ph.D.		
Project Title:	Culture-based Environmental Monitoring of Crop-based Space Food Systems		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Microhost**:Risk of Adverse Health Effects Due to Host-Microorganism Interactions		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Directed Research
Start Date:	12/17/2018	**End Date:**	01/05/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 01/05/2024 per T. Sirmons/NASA JSC. (Ed., 3/2/23) NOTE: End date changed to 12/7/2023 per T. Sirmons/NASA JSC. (Ed., 3/22/22) NOTE: End date changed to 9/30/2022 per PI (Ed., 10/8/21)		
Key Personnel Changes/Previous PI:	Report Updates: added Tanner Hamilton (September 2020 Report), Christian Mena (May 2022 Report) and Brandon Dunbar (May 2022 Report) as co-investigators because of their expertise in processing microbiology surface samples returned from spaceflight.		
COI Name (Institution):	Ott, Mark Ph.D. (NASA Johnson Space Center) Castro, Victoria B.S. (Axiom Space) Hamilton, Tanner B.S. (JES Tech/NASA Johnson Space Center) Mena, Christian (JES Tech/NASA Johnson Space Center) Dunbar, Brandon (GeoControls/NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of the Culture-based Environmental Monitoring of Crop-based Space Food Systems (Veggie Monitoring) investigation is to characterize the microbial community of the Veggie plant production system. This baseline of microorganisms is used to develop microbial requirements for spaceflight-grown produce, and provide inputs to future plant system design. Future work will continue to evaluate the microorganisms that colonize the plant growth system using next generation monitoring technologies to develop future methods for evaluation of produce safety. Of note, the data collected in this study may be used to develop a better understanding of the sources of plant system contamination from the International Space Station (ISS) environment, preflight hardware configuration, water/nutrient supply, plant growth matrix, and the seeds cultivated in the investigation. Twelve surface sampling sessions are required. For each session, four bacterial and fungal samples are collected from locations within the Veggie facility. The samples are collected during flight, and visual enumeration is performed approximately five days post-sampling. The samples are then returned to Earth for culture-based processing and microbe identification. The Veggie hardware surface samples are analyzed, and the microbial data is compared to samples from ISS operational surfaces and the nominal potable water supply.
Rationale for HRP Directed Research:	The MicroHost research plan includes microbial evaluations of food systems and recommendations for spaceflight grown food requirements. Baseline microbial monitoring of the food systems will provide needed data for the development of the microbial requirements. This work can be considered as “highly constrained” since it takes advantage of the operational microbial sampling procedure that is already conducted monthly onboard the ISS, by adding swabbing of the hardware and growing plants and collection of water samples, identical in nature and taken at the same time as the operational sample collections. The highly applied and operational nature of this work makes solicitation or awarding to an external entity not feasible. The MicroHost and Food PRRs (Path to Risk Reduction) identify this work as starting in FY19, and it is preceded by the ground study “Produce Microbiology” (Principal Investigator R. Wheeler) which is underway at Kennedy Space Center and will be completed by the end of FY18, and a Microbial Risk Assessment study initiated in Early FY19. The findings from the characterization of the Veggie system will feed into the Microbial Risk Assessment effort and into the Microbial Requirements Development task, planned to begin in FY20.
Research Impact/Earth Benefits:	Successful production of safe, nutritious food in the challenging conditions in space may contribute to improved food production in harsh and remote environments on Earth.
Task Progress:	Based on the data from the surface samples collected from the Veggie unit, there is a strong correlation between this data and historical data from the surrounding International Space Station (ISS) surfaces. [Ed. Note: The Vegetable Production System (Veggie) is a plant growth unit on the ISS. It is capable of producing salad-type crops to provide the crew with a safe, palatable, and nutritious source of food, while also providing a tool to support recreation and relaxation.] Human commensal and common organisms from the environment were recovered from the returned samples to date. Sites were chosen based on previous results and likelihood of astronaut contact. Access to the interior of the unit is coordinated between Veggie Unit payloads, and as a result, a limited number of samples has been collected. Future sampling will continue to target surface sample collection from within the unit to fully comprehend the impact of microbes on system, plant, and crew health. The results of this effort established the development of a robust microbial community affiliated with the Veggie unit and the similarities to the environmental microorganisms identified on the ISS. Based on this data, it is recommended that the Veggie unit be added to the list of routine air and surface sampling sites for routine microbial monitoring. This addition will allow continual monitoring of microbes present in or on the Veggie environment and identify when additional cleaning of an area may be needed. This work was added to the NASA Kennedy Space Center (KSC) microbial database for continued comparison to historical ISS data as part of the comprehensive evaluation of microbial environmental data. Recommendations: • Add the Veggie system (internal and external) to the nominal ISS microbial monitoring activities • Continue development of the Environmental Microbial Database
Bibliography Type:	Description: (Last Updated:)